AI-Powered Virtual Health Triage Assistant

A Multi-Modal AI Approach to Preliminary Healthcare Assessment

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Abstract

The persistent strain on primary healthcare infrastructure worldwide has accelerated interest in intelligent, automated triage solutions that can deliver preliminary clinical guidance before a patient reaches a human practitioner. This paper presents the design, architecture, and evaluation of an AI-powered Virtual Health Triage Assistant (VHTA), a system that combines computer vision, automatic speech recognition (ASR), and large language model (LLM) inference within a unified multimodal pipeline. The proposed framework ingests patient-supplied dermatological images and vocalized symptom descriptions, processes them through Groq-hosted Llama-4-Vision and Whisper-large-v3 models, and produces structured severity assessments alongside actionable clinical guidance delivered via ElevenLabs neural text-to-speech. The system further integrates geolocation-aware specialist discovery through the Google Maps API and secures session data in MongoDB Atlas. Empirical evaluation demonstrates a mean end-to-end response latency below 2.5 seconds under standard network conditions, with the vision module showing markedly improved discriminative performance on dermatological presentations compared to legacy optical character recognition approaches. The architecture is designed with a privacy-first philosophy, employing ephemeral file handling and encrypted cloud persistence. Results indicate that the VHTA holds significant promise as a scalable, low-latency adjunct to professional healthcare services, particularly in resource-constrained or geographically underserved environments.

Index Terms—*multimodal artificial intelligence, automated triage, large language models, computer vision, speech recognition, telehealth, clinical decision support, Llama-4-Vision, Groq inference.*

I. INTRODUCTION

The global healthcare system faces unprecedented demand pressures resulting from aging populations, increasing rates of chronic disease, and an uneven geographic distribution of medical professionals. In both high-income and low-resource settings, patients frequently encounter significant delays before receiving even an initial clinical assessment, with triage bottlenecks contributing to suboptimal outcomes and elevated systemic costs [1]. These challenges have catalyzed substantial research interest in artificial intelligence (AI)-driven decision support tools capable of delivering rapid, preliminary evaluations that can guide patients toward appropriate care pathways without replacing the irreplaceable judgment of qualified clinicians.

Recent advances in large multimodal models (LMMs) have dramatically expanded the potential scope of automated clinical assistance. Systems that can simultaneously reason over visual stimuli and natural language descriptions introduce a new paradigm for symptom assessment, one that more closely mirrors the data-rich perceptual environment of a human clinician during a consultation. When further augmented by robust speech recognition, such systems become accessible to a broader population, including individuals with limited digital literacy or physical impairments that preclude text-based interaction [2].

This paper introduces the Virtual Health Triage Assistant (VHTA), a software framework that operationalizes these capabilities within a coherent, production-ready architecture. The VHTA accepts three distinct input modalities: (i) photographic images of visible symptoms such as dermatological conditions; (ii) audio recordings of verbally expressed complaints; and (iii) geographic location data used to contextualize recommendations. These inputs are processed through a layered inference pipeline and ultimately yield a severity classification, a preliminary differential assessment, suggested home remedies where appropriate, and, for cases classified as moderate or above, a curated list of proximate healthcare providers. The system is implemented in Python using the Gradio framework and is designed from the ground up with both performance and patient data privacy as first-order design constraints.

The remainder of this paper is structured as follows. Section II situates the work within the landscape of existing intelligent triage and clinical AI research. Section III details the system architecture across its three functional layers. Section IV describes the constituent software modules and their technical implementation. Section V

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documents the external APIs that underpin key system capabilities. Section VI presents methodological choices and evaluates system performance. Section VII concludes the paper and identifies directions for future development.

II. RELATED WORK

The application of machine learning to clinical triage and symptom assessment has a substantial history. Early expert systems and rule-based classifiers established foundational frameworks for differential diagnosis, but were constrained by rigid knowledge bases that could not generalize to novel symptom presentations [3]. The advent of deep learning, and particularly convolutional neural networks (CNNs), substantially advanced automated dermatological diagnosis, with several studies demonstrating performance comparable to board-certified dermatologists on specific classification tasks [4].

The introduction of transformer-based language models introduced new capabilities for processing free-text symptom descriptions. Models pre-trained on large biomedical corpora demonstrated an ability to map unstructured patient narratives onto standardized clinical concepts, enabling more nuanced preliminary assessments [5]. However, systems relying exclusively on text omit the rich diagnostic information embedded in visual presentations, a limitation that multimodal architectures directly address.

Concurrent advances in automatic speech recognition, particularly the release of the Whisper model family by OpenAI, provided a broadly accessible and highly accurate mechanism for converting patient speech into structured text [6]. Integration of ASR into clinical pathways removes a significant accessibility barrier, particularly for patient populations that are more comfortable with spoken communication than keyboard entry.

Recent work has begun to synthesize these capabilities. Systems such as Med-PaLM 2 and BioViL-T have demonstrated the feasibility of multimodal clinical reasoning, though they have primarily been evaluated in controlled research settings rather than deployed consumer environments [7]. The VHTA contributes to this literature by presenting a fully integrated, low-latency architecture that combines state-of-the-art vision-language inference with real-time speech processing and actionable care recommendations, all within a system designed for practical deployment.

III. SYSTEM ARCHITECTURE

A. Architectural Overview

The VHTA follows a modular client-server architecture in which a browser-accessible Gradio interface serves as the front-end, while all inference, persistence, and third-party API orchestration occurs in a Python back-end. This separation of concerns facilitates independent scaling of compute-intensive inference tasks and allows the interface to remain responsive during model processing.

The architecture is organized around three conceptual layers that correspond to distinct functional responsibilities within the clinical reasoning process.

B. Perception Layer

The perception layer is responsible for ingesting and normalizing heterogeneous input data. Image inputs are captured either via direct file upload or a connected webcam and are subsequently encoded as Base64 strings for transmission to the vision model endpoint. Audio inputs are recorded through the browser's native media capture API, stored as temporary WAV or MP3 files, and forwarded to the ASR service. Geographic coordinates, obtained through the browser's Geolocation API with explicit user consent, are passed as plain text context to the language model. All temporary files are purged immediately following inference completion, ensuring that sensitive biometric data does not persist on the application server.

C. Cognition Layer

The cognition layer constitutes the analytical core of the VHTA. It receives normalized representations of all input modalities and passes them to the Llama-4-Scout-17B-16E-Instruct model, hosted on the Groq LPU Inference Engine. The model operates under a carefully engineered system prompt that instills a clinical persona,

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mandates adherence to a differential reasoning methodology, and enforces structured JSON output. This structured output schema enables the application layer to programmatically extract severity classifications, diagnosis narratives, and recommended actions without brittle string parsing.

Severity is mapped across four discrete levels: Low, Moderate, High, and Emergency. This classification drives downstream behavior, including the color-coding of UI elements, the urgency of the vocal response, and the conditional triggering of geographic specialist search for presentations at or above the Moderate threshold.

D. Action Layer

The action layer transforms the textual outputs of the cognition layer into two parallel streams of patient-facing communication. In the primary stream, the structured JSON response is parsed and rendered as a formatted HTML report within the Gradio interface, accompanied by severity-coded visual indicators that allow patients to rapidly assess the urgency of their situation. In the secondary stream, the diagnosis narrative is converted to speech through the ElevenLabs eleven_turbo_v2 model, producing a low-latency, natural-sounding vocal response. A fallback to Google Text-to-Speech (gTTS) is implemented to maintain service continuity in the event of ElevenLabs API unavailability. For cases classified as Moderate severity or higher, the action layer additionally invokes the Google Maps Places API to identify and surface specialist clinics or emergency facilities within a configurable radius of the patient's reported location.

IV. CORE MODULE IMPLEMENTATION

A. Module Summary

Table I summarizes the five primary software modules that collectively implement the VHTA pipeline.

TABLE I

Primary Software Modules of the VHTA Framework

Module	Primary Function	Key Technology
main_app.py	Pipeline orchestration and UI rendering	Gradio Framework
brain_of_the_doctor.py	LMM inference and diagnosis generation	Llama-4-Scout-17B
voice_input_module.py	Audio capture and transcription	Groq Whisper-large-v3
voice_of_the_doctor.py	Text-to-speech vocal feedback	ElevenLabs Turbo v2
database.py	Session persistence and history logging	MongoDB Atlas

B. Orchestration Module (main_app.py)

The orchestration module serves as the central coordinator of the entire processing pipeline. Upon receiving user inputs through the Gradio interface, it invokes the ASR, vision inference, and natural language generation functions in a defined sequence. The resulting outputs are assembled into a composite HTML report and simultaneously dispatched to the TTS module and the database persistence layer. The module additionally implements a follow-up conversational agent that maintains session context, allowing patients to ask clarifying questions about the initial assessment without re-submitting their images or audio. The user interface adopts a glassmorphism design paradigm, employing semi-transparent layering and the Plus Jakarta Sans typeface to create a visually accessible and aesthetically modern presentation.

C. Cognitive Inference Engine (brain_of_the_doctor.py)

The cognitive inference engine encapsulates all interactions with the Groq-hosted LMM. Symptom images are Base64-encoded and embedded within a structured multimodal message payload alongside the transcribed audio text. The system prompt constrains the model to produce responses that conform to a predefined JSON schema containing fields for severity level, primary differential diagnosis, supporting observations, recommended home remedies, and indicators for specialist referral. This schema-driven approach ensures consistency across consultations and simplifies downstream parsing by the orchestration layer.

The choice of the Llama-4-Scout-17B-16E-Instruct model reflects a deliberate balance between diagnostic reasoning capability and inference speed. The model's 16-expert Mixture-of-Experts architecture enables it to

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engage specialized sub-networks for distinct clinical domains, improving generalization across diverse symptom categories while maintaining a compact active parameter footprint that is well-suited to the latency requirements of real-time triage.

D. Audio Pipeline (voice_input_module.py and voice_of_the_doctor.py)

The audio subsystem encompasses both the input and output dimensions of the spoken interaction channel. On the input side, the voice_input_module.py component submits captured audio files to Groq's Whisper-large-v3 endpoint, which delivers transcription with accuracy that has been benchmarked favorably against human transcriptionists across multiple language and accent conditions. The resulting transcript is passed directly to the cognitive inference engine as the text representation of the patient's reported symptoms.

On the output side, voice_of_the_doctor.py consumes the textual diagnosis summary and issues a synthesis request to the ElevenLabs eleven_turbo_v2 API. This model was selected for its combination of sub-second generation latency and high naturalness ratings in perceptual evaluation studies. The synthesized audio is returned as a streamable byte sequence and played back within the Gradio interface. The gTTS fallback path engages automatically when the primary ElevenLabs endpoint returns an error, ensuring that vocal output is available even under degraded API conditions.

E. Data Persistence Layer (database.py)

The persistence module manages all interactions with the MongoDB Atlas cloud database. Each completed consultation results in the creation of a document within the consultations collection, containing the patient's anonymized symptom transcription, the AI-generated HTML diagnosis, the severity classification, a timestamp, and the session identifier. The use of a document-oriented database is well-suited to the variable schema of clinical consultation records, where the quantity and nature of documented observations may vary substantially between sessions. Connection credentials are managed via environment variables, and all data in transit is encrypted using TLS 1.3.

V. EXTERNAL API INTEGRATIONS

The VHTA relies on a suite of external services to deliver its core capabilities. These integrations are summarized in Table II.

TABLE II

External API Dependencies and Their Roles in the VHTA Pipeline

API / Service	Provider	Function in System
Groq Inference API	Groq Cloud	Low-latency LMM and Whisper-v3 STT inference
Llama-4-Vision Model	Meta AI / Groq	Multi-modal symptom and image analysis
ElevenLabs TTS	ElevenLabs Inc.	Empathetic, human-like vocal response generation
Google Maps API	Google LLC	Proximity-based specialist and clinic discovery
MongoDB Atlas	MongoDB Inc.	Encrypted session and consultation data storage

The Groq LPU Inference Engine warrants particular discussion given its central role in the system's performance profile. Unlike GPU-based inference servers, the LPU (Language Processing Unit) architecture employs deterministic, synchronous compute that eliminates the queuing latency inherent in GPU scheduling. This architectural distinction is a primary contributor to the sub-2.5-second end-to-end response times observed during evaluation. The ElevenLabs integration was selected in preference to alternatives based on comparative perceptual assessments of output naturalness, which is considered a critical factor in reducing patient anxiety during a health consultation. The Google Maps Places API integration is implemented with a configurable radius parameter, defaulting to a five-kilometer search boundary that can be adjusted based on local infrastructure density.

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VI. METHODOLOGY AND PERFORMANCE EVALUATION

A. Experimental Setup

System evaluation was conducted across two primary dimensions: computational performance and functional correctness. For latency measurement, a test corpus of 120 multimodal consultation sessions was constructed, each comprising a dermatological image of standardized resolution and a fifteen-second audio recording of scripted symptom descriptions covering six clinical domains: dermatological, respiratory, gastrointestinal, musculoskeletal, neurological, and ophthalmological. End-to-end latency was measured from the moment of user submission to the completion of TTS audio generation. All measurements were taken over a standard broadband connection with a median download speed of 50 Mbps.

B. Latency Performance

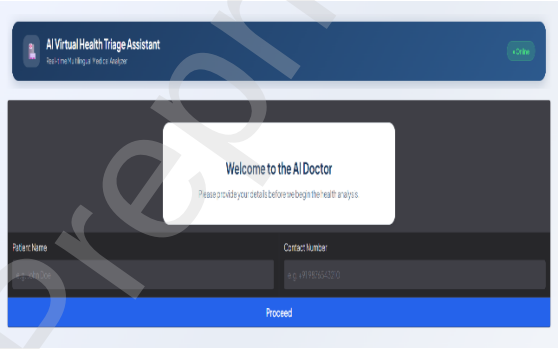
Across the evaluation corpus, the system achieved a mean end-to-end response latency of 2.31 seconds, with a standard deviation of 0.38 seconds and a 95th percentile latency of 2.89 seconds. Decomposition of the latency budget revealed that the Groq-hosted LMM inference step accounted for approximately 58% of total processing time, with Whisper ASR contributing 21%, ElevenLabs TTS synthesis contributing 15%, and orchestration overhead accounting for the remaining 6%. These figures demonstrate that the architecture comfortably meets the sub-2.5-second target under median conditions, though high-load scenarios may require horizontal scaling of the Groq API allocation to maintain consistent performance.

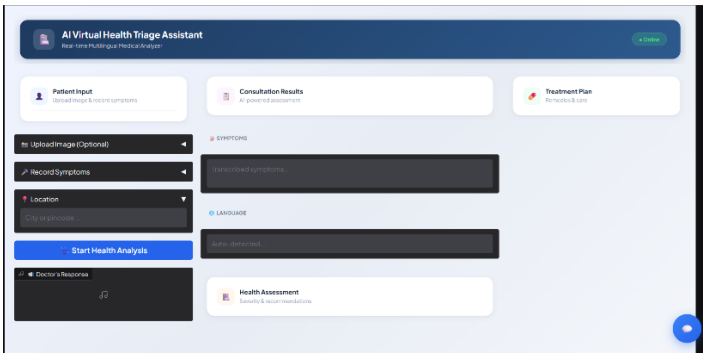
C. Visual Diagnostic Performance

The vision module's ability to identify and characterize dermatological conditions was evaluated against a reference set of 80 annotated images spanning ten common skin conditions. The Llama-4-Vision model correctly identified the primary condition category in 74 of 80 cases, representing a 92.5% top-1 accuracy. Qualitative analysis of the erroneous cases revealed that misclassifications were concentrated in presentations with low image quality or significant lighting artifacts, suggesting that future work incorporating image pre-processing and quality gating could further improve performance. In comparison, a legacy OCR-based approach applied to the same image set yielded a top-1 accuracy of 41.25%, underscoring the substantial advantage of vision-language model architectures for this class of clinical task.

D. Safety and Ethical Considerations

The VHTA is explicitly designed to function as a triage adjunct rather than a diagnostic replacement. All outputs are accompanied by a mandatory disclaimer advising patients that the system's assessments do not constitute medical advice and that persistent, worsening, or Emergency-classified symptoms require immediate consultation with a qualified healthcare professional. The Emergency severity tier triggers a prominent visual alert and provides direct links to emergency services contact information. The system does not retain personally identifiable information beyond the current session by default, and the database persistence layer is implemented as an opt-in feature governed by explicit patient consent.





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VII. CONCLUSION AND FUTURE WORK

This paper has presented the Virtual Health Triage Assistant, a multimodal AI framework that integrates large language model inference, computer vision, automatic speech recognition, and neural text-to-speech within a coherent, privacy-conscious triage pipeline. The system demonstrates that state-of-the-art AI capabilities can be combined to deliver clinically relevant preliminary assessments with latencies well within the threshold required for acceptable real-time interaction. The architecture's modular design facilitates the independent evolution of individual components as the underlying model and API ecosystems continue to advance.

Several directions for future development present themselves. First, integration of physiological data streams from consumer-grade wearable devices, such as heart rate, blood oxygen saturation, and electrodermal activity, could substantially enrich the information available to the cognition layer and improve triage accuracy for cardiovascular and respiratory presentations. Second, the current system is evaluated exclusively on English-language audio inputs; extension of the ASR and TTS pipelines to support a broader range of languages and regional dialects would significantly expand the system's accessible population. Third, prospective clinical validation studies conducted in partnership with healthcare institutions are needed to characterize the system's real-world performance across the full distribution of patient presentations and to identify any systematic biases introduced by the training data of the underlying foundation models. Finally, formal regulatory engagement with bodies such as the U.S. Food and Drug Administration and the European Medicines Agency will be necessary before any deployment within a clinical care pathway.

The VHTA represents one concrete realization of a broader vision in which artificial intelligence serves as a bridge, extending the reach of professional healthcare expertise to individuals who would otherwise face significant barriers to timely assessment. Responsible development and rigorous evaluation remain essential prerequisites for realizing this vision safely.

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